DOPAMINE

High-alert drug
Extravasation necrosis

CHEMICAL NAME 4-(2-aminoethyl)benzene-1,2-diol (hydrochloride)
CLASS Catecholamine inotrope/vasopressor (dose-dependent receptor effects)
TRADE NAMES Intropin · Dopamine HCl infusion · Dopastat

THERAPEUTIC USES / INDICATIONS

- Haemodynamic support in shock (cardiogenic, septic — second-line to noradrenaline).
- Symptomatic bradycardia unresponsive to atropine (alternative to pacing).
- Neonatal/paediatric hypotension (some units).

ADMINISTRATION

- **IV infusion** via central line preferred; syringe pump.
- Do not mix with alkaline solutions (sodium bicarbonate) — inactivated.
- Taper gradually.

DOSAGES

- **Low (2–5 mcg/kg/min):** dopaminergic (renal/mesenteric vasodilation — no proven renal protection).
- **Medium (5–10 mcg/kg/min):** β1 — ↑ contractility and HR.
- **High (>10 mcg/kg/min):** α1 — vasoconstriction.
- **Bradycardia:** 5–20 mcg/kg/min; max ~20 mcg/kg/min.

PHARMACOKINETICS

- **Onset:** 5 min. **Duration:** <10 min after stopping.
- Metabolised by MAO and COMT; ~25% converted to noradrenaline.
- Excreted in urine. **t½:** 2 min.

PHARMACODYNAMICS

- Stimulates **D1** (vasodilation), **β1** (inotropy, chronotropy) and **α1** (vasoconstriction) receptors depending on dose.
- Releases noradrenaline from nerve endings.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tachycardia, palpitations.
- Nausea, vomiting, headache.
- Anxiety.

ADVERSE EFFECTS (SERIOUS)

- Tachyarrhythmias (more than noradrenaline in septic shock).
- Myocardial ischaemia.
- Extravasation necrosis (treat with phentolamine).
- Peripheral ischaemia, gangrene (high doses).

CONTRAINDICATIONS & CAUTIONS

- Phaeochromocytoma.
- Uncorrected tachyarrhythmias or VF.
- **Caution:** hypovolaemia, MAOI use, peripheral vascular disease.

KEY INTERACTIONS

- **MAOIs:** severe hypertension (reduce dopamine dose to 10%).
- **Phenytoin (IV):** hypotension, bradycardia.
- **Halogenated anaesthetics:** arrhythmias.

NURSING MANAGEMENT

- Continuous ECG and BP (arterial line); urine output hourly.
- Check infusion site frequently; have phentolamine available.
- Correct hypovolaemia before starting.
- Monitor for arrhythmias and chest pain.
- Wean slowly.

PATIENT EDUCATION

- Explain monitoring and why the drip rate changes.

KEY POINTS

- Dose-dependent D1/β1/α1 effects.
- More arrhythmias.
- Second-line to noradrenaline in septic shock.
- Central line; phentolamine for extravasation.

SOURCES CHECKED **[380]** Dopamine hydrochloride 200mg/5ml concentrate for solution for infusion ampoules (40mg/ml) (UK SmPC) **[381]** Evans et al., Intensive Care Med 2021 **[382]** Dopamine Hydrochloride, IV (US label)
Full references p. 432–457